

11 (12) September, 2024

Data Encryption Standard (more details)

(mostly taken from William Stallings, *Cryptography and Network Security: Principles and Practices*)

Data Encryption Standard

DES key size is 56 bits. Actually 64, but 8 bits are parity check bits.

The DES algorithm can be summarized as follows:

$$IP(x) = L_0 R_0$$

$$L_i = R_{i-1}$$

$$R_i = L_{i-1} \text{ XOR } f(R_{i-1}, K_i)$$

$$y = IP^{-1}(R_{16} L_{16})$$

IP means initial permutation.

IP (example) →

58	50	42	34	26	18	10	2
60	52	44	36	28	20	12	4
62	54	46	38	30	22	14	6
64	56	48	40	32	24	16	8
57	49	41	33	25	17	9	1
59	51	43	35	27	19	11	3
61	53	45	37	29	21	13	5
63	55	47	39	31	23	15	7

- This table specifies the input permutation on a 64-bit block.
- The meaning is as follows: the first bit of the output is taken from the 58th bit of the input; the second bit from the 50th bit, and so on, with the last bit of the output taken from the 7th bit of the input.
- This information is presented as a table for ease of presentation, it is a vector, not a matrix

Note that, as usual:

$$R_{16} = L_{15} \text{ XOR } f(R_{15}, K_{16})$$

$$L_{16} = R_{15}$$

- But they are switched in the pre-output

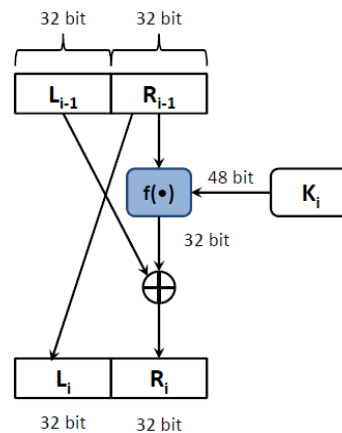
Final Permutation (i.e., IP^{-1})

40	8	48	16	56	24	64	32
39	7	47	15	55	23	63	31
38	6	46	14	54	22	62	30
37	5	45	13	53	21	61	29
36	4	44	12	52	20	60	28
35	3	43	11	51	19	59	27
34	2	42	10	50	18	58	26
33	1	41	9	49	17	57	25

The final permutation is the inverse of the initial permutation; the table is interpreted similarly (i.e., similar to IP):

- That is, the output of the Final Permutation has bit 40 of the pre-output block as its first bit, bit 8 as its second bit, and so on, until bit 25 of the pre-output block is the last bit of the output.

DES Round $i \rightarrow$

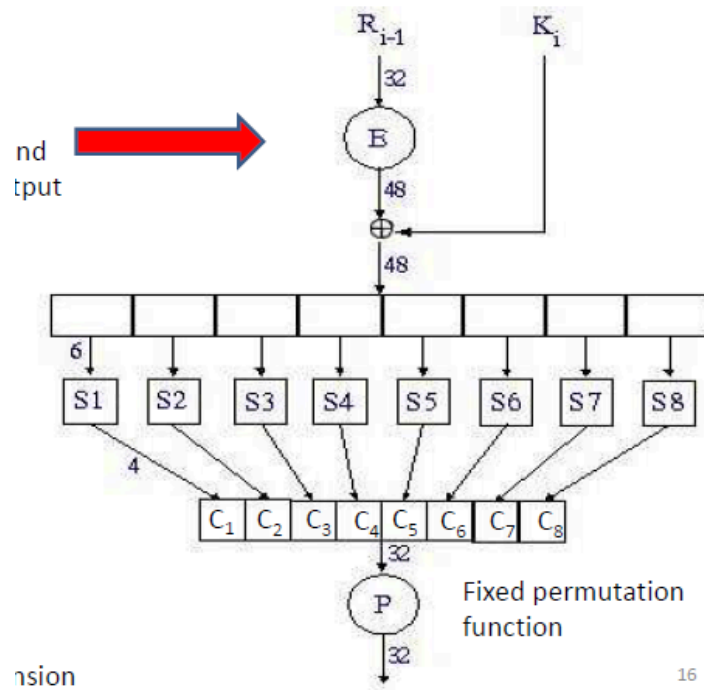


$$L_i = R_{i-1}$$

$$R_i = L_{i-1} \text{ XOR } f(R_{i-1}, K_i)$$

DES $f()$ function $\rightarrow$

E is an expansion function which takes a block of 32 bits as input and produces a block of 48 bits as output.

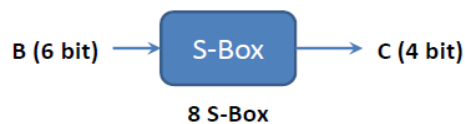


In the table below, 16 bits appear twice, in the expansion (allowing for 32 bits to be expanded to 48 bits)

32	1	2	3	4	5
4	5	6	7	8	9
8	9	10	11	12	13
12	13	14	15	16	17
16	17	18	19	20	21
20	21	22	23	24	25
24	25	26	27	28	29
28	29	30	31	32	1

S-boxes → S-boxes are the only non-linear elements in DES design.

Each of the unique selection functions $S_1, S_2, \dots, S_8$, takes a 6-bit block as input and yields a 4-bit block as output.



- S = matrix 4×16 , values from 0 to 15
- B (6 bit long) = $b_1 b_2 b_3 b_4 b_5 b_6$

- $b_1b_6 = r = \text{row}$ (2 bits: 0, 1, 2, 3)
- $b_2b_3b_4b_5 = c = \text{column}$ (4 bits: 0 to 15)
- C (4 bit long) = Binary representation of $S(r, c)$

Example S_1

Row #	S_1	1	2	3	...											15	Column #
0		14	4	13	1	2	15	11	8	3	10	6	12	5	9	0	7
1		0	15	7	4	14	2	13	1	10	6	12	11	9	5	3	8
2		4	1	14	8	13	6	2	11	15	12	9	7	3	10	5	0
3		15	12	8	2	4	9	1	7	5	11	3	14	10	0	6	13

$S(i, j) < 16$, can be represented with 4 bits

Example: $B = 101111$

$b_1b_6 = r = 11 = \text{row } 3$

$b_2b_3b_4b_5 = c = 0111 = \text{column } 7$

$S(r, c) = 7 \rightarrow \text{Binary} \rightarrow 0111$ (this is our output C_i)

Another Example: $B = 011011$

$b_1b_6 = r = 01 = \text{row } 1$

$b_2b_3b_4b_5 = c = 1101 = \text{column } 13$

$S(r, c) = 5 \rightarrow \text{Binary} \rightarrow 0101$ (this is our output C_i)